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RESEARCH ARTICLE

Integrating Loading Temporal Characteristics and History Into Strain-Adaptive Bone Remodeling for Improved Prediction of Tibial Peri-Prosthetic Bone Mineral Density

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ABSTRACT Accurate prediction of post-implantation bone remodeling is essential for preserving bone stock, mitigating aseptic loosening, and guiding implant design. Classical strain-adaptive bone remodeling models, derived from Wolff's law, relate bone density solely to local strain energy per unit mass. However, *in-vivo* evidence shows that bone adaptation is also regulated by loading frequency, number of loading cycles, and rest periods between loading sessions, factors that strongly influence fluid-induced shear stresses, cellular desensitization, and mechanosensitivity recovery. In this study, a modified strain-adaptive remodeling model that explicitly incorporates these load-related parameters was developed. The model was implemented on a tibia reconstructed from computed tomography data and implanted with a tibial tray. A time-varying loading protocol was defined to mimic post-operative recovery and progressive activity restoration. Bone mineral density (BMD) changes were quantified in regions of interest (ROIs) within the medial and lateral condyles and distal to the implant stem. At two years post-implantation, the modified model predicted BMD reductions of 13.33%, 11.81%, and 1.13% in the medial, lateral, and distal ROIs, respectively, versus 38.09%, 35.77%, and 0.85% predicted by the original formulation. The modified algorithm reproduced both the magnitude and temporal evolution of BMD changes in closer agreement with clinical observations, whereas the classical strain-adaptive model substantially overestimated bone resorption and failed to capture observed remodeling trends. Incorporating load-related parameters into strain-adaptive bone remodeling and accounting for loading history markedly improves biological realism and predictive accuracy, supporting the proposed framework potential application in post-implantation bone remodeling analysis and implant design optimization.

INDEX TERMS Strain-adaptive bone remodeling, strain energy density, loading frequency, number of loading cycles, rest periods, total knee replacement.

I. INTRODUCTION

Bone plays a fundamental role in supporting body movement and maintaining structural stability, and is therefore continuously subjected to a wide range of dynamic forces during daily activities. Its mechanical competence arises from a unique composite structure that confers strength, stiffness, and toughness. Bone consists of approximately 60% minerals, 30% organic extracellular matrix, and 10% cells [1]. The mineral phase provides stiffness and strength [2], whereas the organic matrix contributes to toughness [3], allowing bone to withstand complex multiaxial loading conditions.

Mechanical demands on bone vary over time due to lifestyle changes, aging, injury, and surgical intervention. Bone possesses a remarkable capacity to adapt its structure

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to these changing demands through continuous remodeling driven by cellular activity [4]. This adaptive behavior is classically described by Wolff's law [5], which states that bone architecture is regulated by the mechanical stresses it experiences, resulting in an optimized structure suited to its functional requirements. Experimental evidence supporting this principle was provided by Turner et al. [6], who applied daily cyclic four-point bending loads of varying amplitudes to the tibiae of adult rats and showed that endocortical mineral apposition rates increased with loading amplitude. Conversely, reduced mechanical loading leads to bone loss, as observed in astronauts exposed to prolonged weightlessness [7], and in patients undergoing orthopaedic implantation [8], [9], where stiffness mismatches between implants and host bone result in stress-shielding-driven bone resorption.

Building on Wolff's law, Huiskes et al. [10] developed a strain-adaptive bone remodeling algorithm in which changes in strain energy per unit mass from a reference state serve as the mechanical stimulus for density adaptation. In this framework, increases in strain energy initiate bone formation, whereas reductions lead to bone resorption. This algorithm has been extensively employed to assess the influence of implant parameters such as design [11], fixation technique [12], and material [13] on bone remodeling, and has been coupled with topology optimization schemes to enhance implant performance [14], [15].

Although strain-adaptive remodeling theories are well-established, a more comprehensive understanding of how bone remodels in response to changes in its mechanical environment is essential for accurately assessing and improving implant performance. It is widely accepted that shear stresses on cell membranes, generated from deformation-induced interstitial fluid flow within bone tissue, constitute the primary mechanical stimulus for bone cells [16]. Importantly, interstitial fluid flow arises only under cyclic loading [17]. This was demonstrated *in-vivo* by Robling et al. [18], who subjected separate groups of rats to either dynamic or static compressive loading on the ulna over a two-week period. Dynamic loading significantly enhanced bone formation on both periosteal and endocortical surfaces, whereas static loading failed to stimulate bone formation.

Beyond the necessity of cyclic loading, loading frequency also influences the magnitude of fluid-induced shear stresses. Turner et al. [19] investigated this effect using a rat tibial bending model and reported that bone formation rates increased linearly with loading frequency between 0.5 and 2 Hz. Based on these findings, Turner [20] proposed that the mechanical stimulus driving bone adaptation—defined as a scalar measure of the local mechanical signal sensed by bone cells—is proportional to the product of strain magnitude and loading frequency. This relationship successfully reproduced experimental observations in [6] and [19] and predicts a vanishing stimulus under static loading conditions, providing an explanation for the absence of bone formation observed in the static loading group in [18].

Loading duration, commonly expressed as the number of loading cycles, also impacts bone adaptation. Rubin and Lanyon [21] revealed that increasing daily compressive loading cycles in isolated turkey ulnae enhanced bone mineral content; however, the response saturated rapidly, with minimal additional gains beyond 36 cycles. Similar results were reported by Umemura et al. [22] in a rat jumping model, where increases in tibial bone mass plateaued despite increases in daily loading events. Owing to diminishing returns with increasing cycle number, bone response exhibits a logarithmic dependence. Turner [20] incorporated both frequency (f) and cycle count (N) into an effective mechanical stimulus (E), expressed as:

$$E \propto \sum_{i=1}^{n} \epsilon_i f_i \log(1+N_i) \tag{1}$$

where ϵ_i is the effective strain under loading case i and n is the number of loading cases considered. Notably, similar dependencies on strain magnitude, frequency, and duration have also been observed *in-vitro* [23], where oscillatory fluid flow has been shown to initiate osteogenic responses in bone mesenchymal stem cells.

Rest periods between loading sessions also play a vital role in regulating bone mechanosensitivity. Continuous loading leads to cellular desensitization, whereas rest periods allow recovery of mechanosensitivity. This is supported by the findings of Robling et al. [24], who demonstrated that introducing rest intervals between loading bouts in a rat tibial bending model significantly enhanced bone formation, with longer rest periods producing greater osteogenic responses. Based on these observations, the recovery of mechanosensitivity has been described as:

$$\text{Recovery}(\%) = 100(1 - e^{-\frac{t}{\tau}}) \tag{2}$$

where t is the rest duration and τ is a constant equivalent to 6 hours [25]. Collectively, these experimental findings demonstrate that bone adaptation is governed not only by loading magnitude or strain energy, but also by loading frequency, duration, and rest periods. Because classical strain-adaptive bone remodeling algorithms consider only changes in strain energy as the governing stimulus, their ability to accurately predict time-dependent bone adaptation is inherently limited, potentially leading to inaccurate assessments of peri-prosthetic bone loss and implant performance. Therefore, the objective of this study is to modify the strain-adaptive bone remodeling algorithm by incorporating loading frequency, number of cycles, and rest periods into the mechanical stimulus, and to evaluate its predictive capability against clinical tibial post-implantation data. We hypothesize that the modified, load-history-dependent model will provide improved agreement with clinically observed temporal changes in tibial bone density compared to the original formulation, thereby enabling more reliable prediction of implant-induced bone loss and supporting improved implant design and rehabilitation strategies.

II. METHODOLOGY

A. MODIFIED STRAIN-ADAPTIVE BONE REMODELING ALGORITHM

The original strain-adaptive bone remodeling algorithm is extended to explicitly account for load-related parameters that influence bone adaptation by redefining the mechanical stimulus driving remodeling. In the original formulation, the evolution of bone apparent density (ρ) is governed by changes in strain energy per unit mass ($S = \frac{1}{2}\frac{\sigma:\epsilon}{\rho}$), which serve as the mechanical stimulus controlling the rate of density change ($\frac{d\rho}{dt}$) [10]. The remodeling law is expressed as:

$$\frac{d\rho}{dt} = \begin{cases} cA(\rho)\left[S_{post} - (1-s)S_{pre}\right], S_{post} < (1-s)S_{pre} \\ 0, (1-s)S_{pre} \le S_{post} \le (1+s)S_{pre} \\ cA(\rho)\left[S_{post} - (1+s)S_{pre}\right], S_{post} > (1+s)S_{pre} \end{cases} \quad (3)$$

where S_{pre} and S_{post} denote the pre- and post-operative average strain energies per unit mass, respectively, and c is a remodeling coefficient that governs the sensitivity of the density evolution to deviations from the reference state S_{pre} [26]. As shown in Figure 1, remodeling occurs only when the difference between S_{post} and S_{pre} exceeds a non-dimensional threshold s, defining a "dead zone" around the reference state within which no adaptation takes place. Outside this region, the magnitude of remodeling rate is further modulated by the specific surface area available for remodeling, $A(\rho)$, expressed as function of ρ [27].

To incorporate the temporal characteristics of loading, an effective strain energy is defined to account for loading frequency, number of loading cycles and rest periods, following the methodology proposed in [28]. Strain energy per unit mass was retained as the core mechanical quantity to preserve the mechanobiological basis of the original model, while load-related terms were introduced as multiplicative modifiers of this quantity. To maintain dimensional consistency, loading frequency was normalized with respect to an activity-specific reference frequency f_o, which represents the frequency of the loading cycle of an activity performed at normal speed (e.g. f_o of gait cycle at normal walking speed is around 0.833 Hz). As will be discussed in section B, the usage of aid tools post-surgery reduces activity speed and results in loading frequencies lower than f_o. Multiple loading cases were superimposed to represent activities and their relative contribution over a day. This allows differences in activity level (i.e. number of loading sessions) between pre- and post-operative conditions to be captured explicitly. The resulting effective strain energy is defined as:

$$S_{eff} = k\sum_{i=1}^{n} S_i\left(\frac{f_i}{f_{o_i}}\right)\log(1+N_i)\left(1-e^{-\frac{t}{6}}\right) \quad (4)$$

where S_i is the strain energy per unit mass under loading case i, f_i is the loading frequency, N_i is the number of loading cycles, and t_i is the rest duration (in hours) that precedes the loading session. The term $\log(1+N_i)$ captures the experimentally observed saturation of the osteogenic response with increasing cycle number [20], while $(1-e^{-\frac{t_i}{6}})$ represents the recovery of mechanosensitivity during rest periods [25]. The constant k is a proportionality factor proposed by Turner [20]. The effective strain energy is then embedded into the original remodeling law (3) by replacing S_{pre} and S_{post} with their effective counterparts, $S_{eff_{pre}}$ and $S_{eff_{post}}$. As a result, the product ck appears in the remodeling law, but k is absorbed into the remodeling coefficient c and is calibrated as a single parameter (c) in the numerical implementation. In this model, remodeling can be triggered by post-operative changes in any load-related parameter—frequency, cycle number, or rest period—even if the strain energy remains unchanged. This extension preserves the structure of the classical strain-adaptive framework while enabling a more physiologically realistic description of time-dependent bone adaptation under altered activity patterns following implantation.

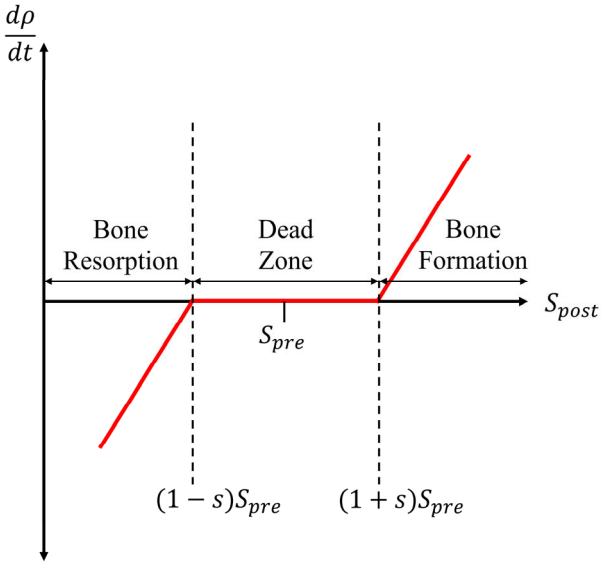


FIGURE 1. Graphical representation of the strain-adaptive bone remodeling algorithm linking the rate of change in bone density ($\frac{d\rho}{dt}$) to the change in average strain energy per unit mass in bone between post-implantation (S_{post}) and pre-implantation (S_{pre}) phases. Bone adaptation takes place in regions outside the dead zone, whose range is controlled by the threshold s. Bone resorption occurs in underloaded bone ($S_{post} < (1-s)S_{pre}$) while bone formation occurs in overloaded bone ($S_{post} > (1+s)S_{pre}$).

Following the formulation of the modified algorithm, the model parameters governing adaptation are identified and calibrated against experimental and clinical data. The threshold s, which controls the width of the dead-zone around S_{pre}, remains uncertain and is known to vary with individual metabolic status and age [29]. Values such as $s = 0.5$ [14] and $s = 0.75$ [11] have been widely employed in computational remodeling studies. In the present work, a lower threshold of $s = 0.35$ was adopted, as this value has been shown to yield predictions in good agreement

with clinical observations in both hip [30] and knee [31] arthroplasty.

The remodeling coefficient c, which scales the rate of density change per unit stimulus, is calibrated to reproduce physiologically realistic tibial adaptation rates. Previously reported values of 129.6 [26] and 32.4 g/[cm^3.mm^2.(J/g).month] [32] were derived using relatively coarse finite element meshes and non-physiological constraints on density change and therefore are not directly adopted. Instead, clinical data were used to define plausible bounds. Leblanc et al. [33] reported an average tibial bone mineral loss rate of 0.56%/month after 17 weeks of bed rest, whereas Andersson and Nilsson [34] observed losses of up to 6.5%/month in the proximal tibia following ligament injury and non-weight-bearing immobilization. As knee arthroplasty patients are mobilized immediately after surgery, the latter value (6.5%/month) is taken as an upper bound on the magnitude of density loss under reduced loading.

To relate bone mineral content (BMC) loss to ρ, the relationships reported in [35] were employed. Bone tissue density ρ_{tissue} (density excluding porosity) was shown to depend approximately linearly on BMC, while the relationship between ρ and ρ_{tissue} varies with ρ. For cortical bone, however, ρ_{tissue} can be approximated as a linear function of ρ. Hence, a 6.5% reduction in BMC can be taken to correspond to a 6.5% reduction in ρ. Assuming a maximum cortical density of 2 g/cm^3 [35], this yields an upper bound on the local rate of density change of 0.13 g/cm^3/month. The coefficient c was therefore calibrated such that the maximum rate of density change at any integration point during the first computer time unit—the period in which the highest remodeling rate occurs—does not exceed this bound, specifically:

$$c \leq \frac{0.13}{\max[A(\rho)(S_{eff_{post}} - (1 \pm s)S_{eff_{pre}}]} \tag{5}$$

where the denominator represents the maximum (in magnitude) value of the product $A(\rho)(S_{eff_{post}} - (1 \pm s)\, S_{eff_{pre}})$ over the tibial domain over both the resorption $(1 - s)$ and formation $(1 + s)$ branches. This calibration anchors the remodeling rate to clinically observed extremes of disuse-related bone loss.

Finally, distinct remodeling coefficients were introduced for bone resorption and formation to reflect their asymmetric kinetics. Clinical and experimental evidence indicates that bone formation proceeds substantially more slowly than resorption. Following bed rest in [33], tibial bone recovery occurred at approximately half the rate of loss, and an exponential recovery model suggested that spaceflight-induced bone loss during 4-6 month missions may require up to three years for recovery [7]. Moreover, bone loss continued, although at a reduced rate, after remobilization of ligament tear patients [34]. Consistent with the characteristic durations of remodeling cycle—approximately 2–3 weeks for resorption and 4–6 months for formation [36]—the bone formation coefficient was chosen to be 10 times smaller than the resorption coefficient. Incorporating these choices, the final form of the modified strain-adaptive remodeling law is:

$$\frac{d\rho}{dt} = \begin{cases} cA(\rho)\left[S_{eff_{post}} - 0.65\, S_{eff_{pre}}\right], S_{eff_{post}} < 0.65\, S_{eff_{pre}} \\ 0,\ 0.65\, S_{eff_{pre}} \leq S_{eff_{post}} \leq 1.35 S_{eff_{pre}} \\ \frac{c}{10}A(\rho)\left[S_{eff_{post}} - 1.35\, S_{eff_{pre}}\right], S_{eff_{post}} > 1.35\, S_{eff_{pre}} \end{cases} \tag{6}$$

where $S_{eff_{pre}}$ and $S_{eff_{post}}$ are the pre- and post-operative effective strain energies, and the factors 0.65 and 1.35 correspond to $(1 - s)$ and $(1 + s)$ for $s = 0.35$. This formulation preserves the dead zone around the reference state, constrains the maximum remodeling rate to clinically plausible values, and accounts for the slower kinetics of bone formation relative to resorption.

B. LOADING PROTOCOL

The knee joint is subjected to substantial mechanical loading during daily activities such as standing up, sitting down, walking, and stair ambulation. Although these activities contribute to joint loading, only walking and stair activity are considered in this study, as they involve cyclic and repetitive load patterns with definable frequencies and cycle counts. During these activities, the knee joint experiences compressive and shear forces, as well as bending and torsional moments. However, *in-vivo* measurements from instrumented knee implants have shown that peak shear forces are 10-20 times lower than peak compressive forces, with the resultant joint force acting predominantly in the vertical direction. Furthermore, while peak compressive forces exceed 260% of body weight (BW), the associated moments, in Nm, are typically less than 3% BW [37]. On this basis, only compressive loading was modeled in the present analysis.

Computational studies of bone adaptation after total knee replacement (TKR) commonly assume physiological joint loads representative of normal, pain-free daily activities [11], [13], [31], [38]. Such assumptions, however, do not reflect the mechanical environment during the early post-operative period. TKR is generally indicated in patients with severe pain due to knee osteoarthritis [39], which substantially reduces pre-operative mobility. Immediately after surgery, additional pain and swelling further limit function. To alleviate symptoms and protect the operated limb, mobility aids such as walkers and crutches are typically prescribed for approximately 4–6 weeks post-operatively [40], [41], [42], shifting a portion of body weight away from the operated knee and thereby reducing both load magnitude and walking speed. Pain also limits activity duration, and patients are usually encouraged to perform multiple short activity bouts throughout the day during early rehabilitation [43]. As recovery progresses, activity levels are gradually increased [44], and

most patients return to their pre-osteoarthritic pain activity levels within roughly three months post-surgery [45].

Pre-operative activity in the present work was set conservatively: recognizing that osteoarthritis significantly restricts mobility yet avoiding unrealistically low assumptions. Total daily walking and stair activity was limited to 25 minutes, divided into two sessions, representing a typical routine such as commuting to and from work. A conservative, time-dependent loading protocol was then defined to represent successive rehabilitation stages following TKR. During the first two post-operative weeks, walking was performed with a walker, and stair ascent was assisted by crutches and a handrail. Four loading sessions per day were prescribed, each lasting 5 minutes. Over the subsequent two weeks, the walker was replaced by crutches, allowing a gradual increase in joint loading while maintaining the same number and duration of daily sessions. After four weeks—consistent with the typical return-to-work timeframe for patients with sedentary occupations [46]—the pre-operative activity duration (25 minutes/day in two sessions) was resumed, but crutches were still used. After six weeks, all activities were permitted without mobility aids. During the following two weeks, an additional 10-minute activity session was introduced after a 4-hour rest period following work. This extra session was then extended by 5 minutes every two weeks, reaching a maximum of 20 minutes. No further increases in activity duration were considered, reflecting the expectation that patient activity levels plateau rather than increase indefinitely.

Load magnitudes and corresponding activity frequencies were obtained from the OrthoLoad database [47], assuming a body mass of 100 kg. Stair activity was taken to represent a single floor, corresponding to 20 steps. As no *in-vivo* measurements were available for stair descent using mobility aids, this activity was excluded during the first 6 weeks. This is consistent with evidence showing that stair descent is more biomechanically demanding than ascent due to higher eccentric muscular control requirements [48], making it less likely to be routinely performed in early rehabilitation. Each loading session consisted of all activities permitted at the rehabilitation stage. An 8-hour work period was assumed to represent a rest phase for the operated knee. Although this simplification does not capture the full complexity of daily loading patterns, it avoids impractical assumptions regarding step-by-step sequencing and is consistent with *in-vivo* protocols in which normal activity occurs between controlled loading bouts [22], [24]. For each activity, the activity-specific reference frequency (f_o) was defined as the frequency of the loading without mobility aids. These reference values were used to normalize loading frequencies in the effective strain energy definition (4). Figure 2 summarizes the details of the rehabilitation protocol and displays the daily loading schedule in the pre- and post-operative phases, while Table 1 lists the load magnitudes, frequencies, number of load cycles, and rest periods of the activities in the time-varying loading protocol.

Bone remodeling was also predicted using the original strain-adaptive law (3), where time-varying loading was not considered; instead, physiological ''normal'' loading conditions were applied throughout the 2-year adaptation period, consistent with previous studies [12], [31], [38]. In addition to the original and modified remodeling formulations, two supplementary simulations were performed using the modified algorithm to assess the influence of prolonged use of mobility aids, with aid durations of 2 and 3 months modeled over a one-year period. In these scenarios, pre-osteoarthritic-pain activity levels were reached 4 and 6 months post-surgery, respectively.

C. FINITE ELEMENT MODEL

Following the development of the modified strain-adaptive bone remodeling algorithm (6) and the definition of the loading protocol, both were implemented within a finite element (FE) framework to numerically predict tibial bone adaptation. Computed tomography (CT) images of the female subject from the Visible Human Project [49] were imported into Mimics Research 21.0 (Materialise, Leuven, Belgium), where the tibia was segmented using Hounsfield unit (HU)-based thresholding and reconstructed into a three-dimensional geometry after removal of surrounding anatomical structures. The female specimen was selected because the donor age (59 years) falls within the typical age range of patients undergoing TKR and is therefore more representative of the target population than the younger male specimen. The reconstructed tibia was subsequently processed in 3-matic Research 13.0 (Materialise, Leuven, Belgium), where the geometry was smoothed and prepared for virtual implantation A virtual proximal tibial resection was performed at a depth of 8 mm below the lowest point of the highest tibial plateau, consistent with previous TKR modeling studies [12], [38]. A cylindrical cavity was then created using Boolean operations to accommodate the implant stem. The tibia was truncated 11 cm distal to the proximal resection plane and subdivided into two anatomical regions: (i) the proximal tibia, extending 8 cm from the top, and (ii) the diaphyseal region located below the segment. The final tibial geometry therefore consists of three distinct parts: the proximal tibia, the diaphysis, and the bone volume removed during surgery, which comprises both the removed plateau section and the stem cavity.

A custom tibial tray was designed for the present study, inspired by commercially available systems. The overall baseplate geometry resembles that of the U2 Knee System (United Orthopedic Corporation, Taiwan), without directly replicating any proprietary design features. The tray size was selected as the largest configuration that could be positioned on the resected surface while maintaining a maximum overhang less than 2 mm, in line with clinical recommendations [50]. Both the tibial and implant geometries are demonstrated by Figure 3, while the main design parameters of the tibial baseplate are summarized in Table 2.

(a) Rehabilitation timeline

Rehabilitation phase	Time post-surgery	Mobility aid	Daily loading sessions	Total activity duration	Activities in each session
Pre-operation	Before surgery	None	2 sessions	25 min (2 x 12.5)	All (Walking, stair ascending + descending)
Early rehab 1	0-2 weeks	Walker	4 sessions	20 min (4 x 5)	Walking + stair ascending
Early rehab 2	2-4 weeks	Crutches	4 sessions	20 min (4 x 5)	Walking + stair ascending
Transition 1	4-6 weeks	Crutches	2 sessions	25 min (2 x 12.5)	Walking + stair ascending
Transition 2	6-8 weeks	None	2 sessions	25 min (2 x 12.5)	All
Progression 1	8-10 weeks	None	3 sessions	35 min (2 x 12.5) + (1 x 10)	All
Progression 2	10-12 weeks	None	3 sessions	40 min (2 x 12.5) + (1 x 15)	All
Maintenance	12+ weeks	None	3 sessions	45 min (2 x 12.5) + (1 x 20)	All

(b) Daily loading schedule

Time	Morning		Afternoon		Evening	Night
Pre-operation	12.5 min	Rest (Desk job) ~ 8 h	12.5 min	Rest ~ 15.5 h		
0-2 weeks	5 min	Rest ~ 4 h	5 min	Rest ~ 4 h	5 min	Rest ~ 11.5 h
2-4 weeks	5 min	Rest ~ 4 h	5 min	Rest ~ 4 h	5 min	Rest ~ 11.5 h
4-6 weeks	12.5 min	Rest (Desk job) ~ 8 h	12.5 min	Rest ~ 15.5 h		
6-8 weeks	12.5 min	Rest (Desk job) ~ 8 h	12.5 min	Rest ~ 15.5 h		
8-10 weeks	12.5 min	Rest (Desk job) ~ 8 h	12.5 min	Rest ~ 4 h	10 min	Rest ~ 11 h
10-12 weeks	12.5 min	Rest (Desk job) ~ 8 h	12.5 min	Rest ~ 4 h	15 min	Rest ~ 11 h
12+ weeks	12.5 min	Rest (Desk job) ~ 8 h	12.5 min	Rest ~ 4 h	20 min	Rest ~ 11 h

Mobility aids: Walking and stair ascending

Walking, stair ascending and descending

FIGURE 2. Illustration of the time-varying loading protocol. (a) Rehabilitation timeline showing mobility aid usage, activity duration and permitted activities in each rehabilitation phase. (b) Daily loading schedule showing the time and duration of each loading session (block). All activities are performed in each session, except for the period of aid tools where stair descent is avoided.

Accurate representation of material heterogeneity was essential, as strain energy generation within bone is highly sensitive to local stiffness. Elastic properties were therefore assigned by mapping CT Hounsfield unit (HU) values to bone apparent density (ρ) and elastic modulus (E). Although multiple $\rho - \text{HU}$ and $E - \rho$ relationships have been proposed [51], these relationships depend on the anatomic site [52], [53]. In this study, the tibia-specific relationship reported in [52] was employed to convert HU values to apparent density ($\rho = 0.114 + 0.000916 \cdot HU$, g/cm^3). Regions with HU < 0 were assumed to represent bone marrow and were assigned an elastic modulus of 20 MPa [54]. For the proximal tibia, the elastic modulus was calculated from ρ (in g/cm^3) using the relationship in [55] ($E = 4778\rho^{1.99}$, MPa). In the diaphyseal region, cortical bone properties were defined according to the formulation in [56] ($E = 3891\rho^{2.39}$, MPa). The highest elastic modulus assigned exceeded 14 GPa in the cortical diaphysis. The distribution of

TABLE 1. Load, frequency (f), number of loading cycles (N) and rest period that precedes the loading bout (t) of each activity in the time-varying loading protocol (Figure 2), with loading bouts beginning at morning. Each loading bout consists of all activities, except during the period where aid tools are utilized, where no stair descending took place. Frequencies of the loads during the pre-operative period represent f_o in (4).

Time	Loading Bout	t (hours)	Walking			Stair Ascending			Stair Descending		
			Load (N)	f (Hz)	N	Load (N)	f (Hz)	N	Load (N)	f (Hz)	N
Pre-operation	1	15.5	2560	0.83	604	3100	0.8	10	3394	0.8	10
	2	8									
0-2 weeks	1	11.5	1883	0.385	108	2315	0.53	10	-		
	2	4									
	3	4									
	4	4									
2-4 weeks	1	11.5	2138	0.714	201	2315	0.53	10	-		
	2	4									
	3	4									
	4	4									
4-6 weeks	1	15.5	2138	0.714	604	2315	0.53	10	-		
	2	8									
6-8 weeks	1	15.5	2560	0.83	604	3100	0.8	10	3394	0.8	10
	2	8									
8-10 weeks	1	11	2560	0.83	604	3100	0.8	10	3394	0.8	10
	2	8									
	3	4			479						
10-12 weeks	1	11	2560	0.83	604	3100	0.8	10	3394	0.8	10
	2	8									
	3	4			729						
12+ weeks	1	11	2560	0.83	604	3100	0.8	10	3394	0.8	10
	2	8									
	3	4			979						

TABLE 2. Dimensions and design parameters of the custom tibial tray.

Design parameter	Description
Medio-lateral width: w (mm)	69
Antero-posterior length: l (mm)	47
Baseplate thickness: t (mm)	3
Baseplate slope (°)	0
Baseplate design	Symmetric
Stem length: h (mm)	35
Stem design	Cylindrical

the HU and elastic modulus on the external surface of the tibia is demonstrated by Figure 4. As the proximal metaphyseal cortex is thin and lies beneath superficial trabecular bone over much of the surface, the displayed contour predominantly represents cancellous bone rather than cortical bone.

All internal interfaces within the intact tibia model, including those between the proximal region, diaphysis, and removed bone volume, were modeled as fully bonded (tie constraints). As shown in Figure 5a, a 2-mm polymethylmethacrylate (PMMA) cement layer was introduced between the bone and the tibial tray in the tibia–implant construct, with the bone–cement and cement–implant interfaces assumed to be fully bonded. Bone, implant, and bone cement were modeled as linear elastic and isotropic materials. Although bone is intrinsically orthotropic, Peng et al. [54] reported only minor differences between isotropic and orthotropic representations in intact bone models, and Doblaré and García [57] showed that bone anisotropy decreases following implantation, resulting in behavior closer to isotropic. On this basis, an isotropic description was adopted as a reasonable approximation. The material properties of all components are listed in Table 3.

Peak compressive joint loads associated with walking, stair ascent, and descent (Table 1) were applied to the medial and lateral condyles as shown in Figure 5b. Medial-to-lateral load-sharing ratios of 60:40 for walking [58] and 55:45 for stair ambulation [59] were adopted. Peak compressive forces for each activity were applied at the centers of pressures indicated in Figure 5c [38]. The distal end of the tibia was fully constrained in all degrees of freedom. Both the intact and tibia–implant configurations were discretized using linear tetrahedral elements with an average length of approximately 0.7 mm, selected based on mesh convergence analysis of S, with coarser mesh being employed in the diaphysis. Bone remodeling simulations were time-scaled using computer time units (CTUs), where each iteration was taken to represent a physical time span of two weeks, consistent with the temporal resolution of the loading protocol (Table 1) and closely aligned with the time scaling adopted in [30]. After each iteration, bone apparent density and corresponding elastic properties were updated elementwise to reflect remodeling-induced changes. All FE analyses and remodeling simulations were conducted using ABAQUS 2020 (Dassault Systèmes, USA).

To enable quantitative comparison with clinical findings, two-dimensional antero-posterior (AP) projections were generated from the FE results. Element numbers were mapped onto a three-dimensional grid with cubic voxels of

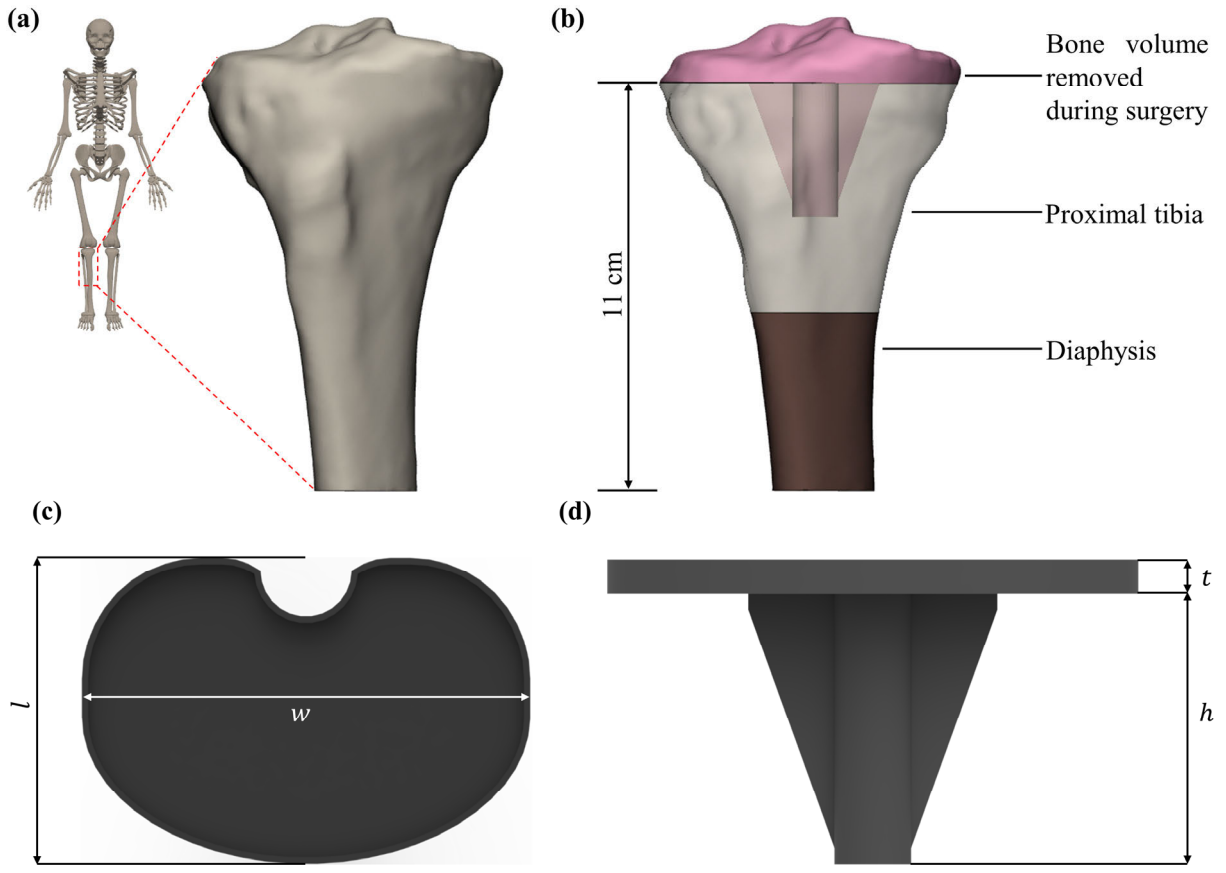


FIGURE 3. (a) Anterior view of the reconstructed intact tibia 3D model. (b) Division of the tibia into three distinct parts. (c) Superior view of the custom tibial tray with medio-lateral width *w* and antero-posterior length *l*. (d) Anterior view of the custom tibial tray with baseplate thickness *t* and stem length *h*. Dimensions of the tibial tray are provided in Table 2.

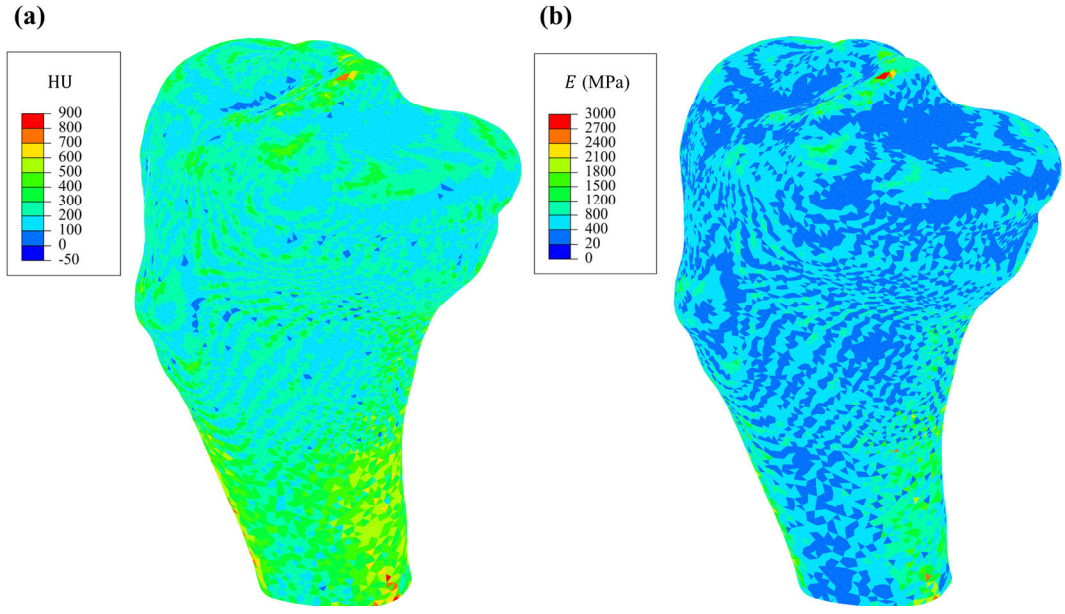


FIGURE 4. Tibial distribution of: (a) Hounsfield units (*HU*) and (b) Elastic modulus (*E*) in MPa.

volume 0.125 mm^3, and the corresponding bone mineral content (BMC) within each voxel was computed. BMC values were then summed along the AP direction, where post-operative remodeling effects are most pronounced,

TABLE 3. Material properties of the components of the finite element models.

Component	Elastic Modulus (MPa)	Poisson's ratio
Proximal tibia [55]	$4778\rho^{1.99}$	0.3
Cortical bone (diaphysis) [56]	$3891\rho^{2.39}$	0.3
Bone marrow [54]	20	0.48
Tibial tray (Ti6Al4V) [13]	110000	0.3
Bone cement (PMMA) [13]	2200	0.3

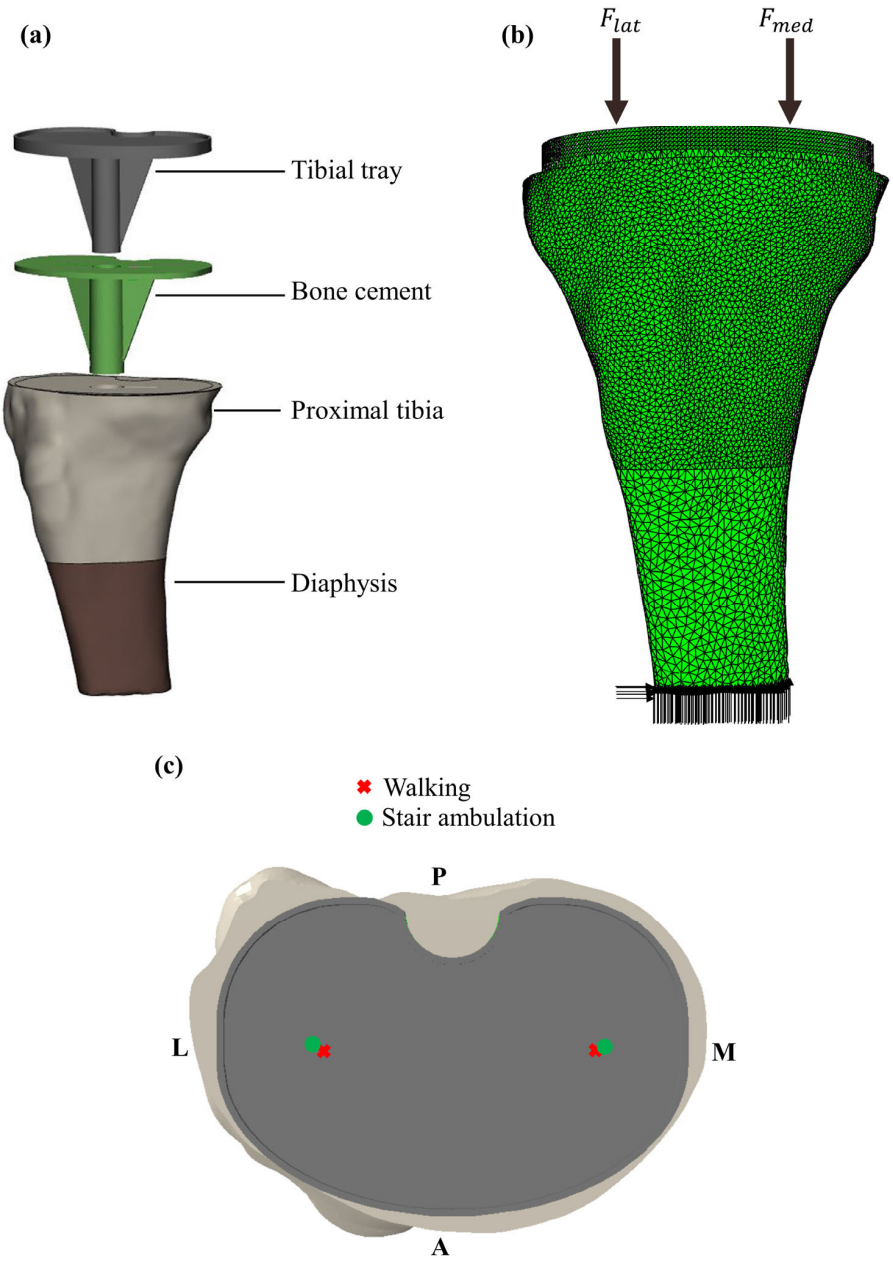


FIGURE 5. (a) Components of the tibia–implant finite element model. (b) Anterior view of the meshed tibia–implant model demonstrating the applied loading and boundary conditions. The peak compressive load is distributed between the medial (F_{med}) and lateral (F_{lat}) condyles in both the intact tibia and tibia–implant models. (c) Superior view of the tibia–implant model showing the medial and lateral centers of pressure for walking and stair ambulation [38]. A: anterior, P: posterior, L: lateral, M: medial.

to generate virtual AP bone mineral density (BMD) maps at successive time points. This approach allow for direct comparison of percentage changes in BMD with published clinical data over a 2-year adaptation period [60], [61],

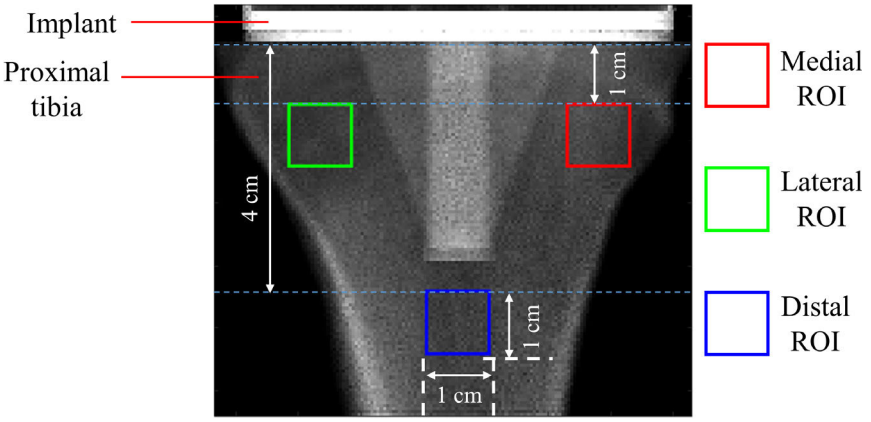


FIGURE 6. Virtual antero-posterior (AP) bone mineral density (BMD) map with the regions of interest (ROIs) defined following the methodology in [69]. The medial and lateral ROIs are located 1 cm beneath the baseplate within the respective condyles, while the distal ROI is positioned 4 cm distal to the baseplate, centered along the medio-lateral position of the stem. Each ROI covers an area of 1 cm^2.

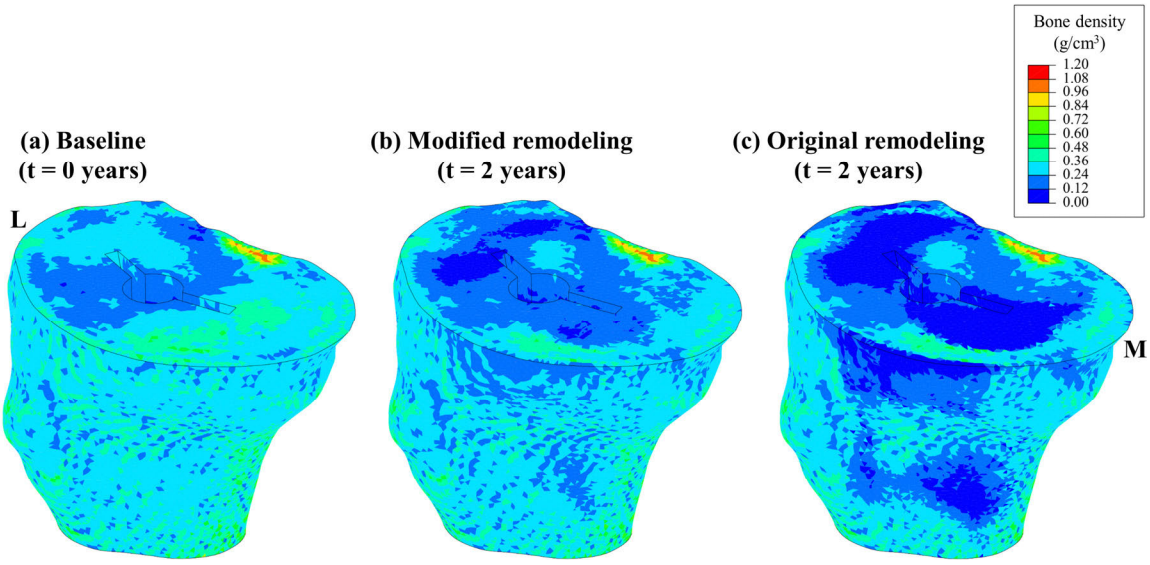


FIGURE 7. Contour plots of bone apparent density of the proximal tibia at (a) immediately post-implantation (t = 0), (b) after 2 years of remodeling using the modified algorithm, and (c) after 2 years using the original remodeling algorithm, which predicts greater extent of bone resorption in the entire proximal tibia. L: lateral. M: medial.

[62], [63], [64], [65]. Although absolute pre-operative BMD values are subject-dependent, Abu-Rajab et al. [66] reported that relative changes in BMD are largely independent of baseline values. The choice of a 2-year remodeling window is supported by single-photon emission CT studies indicating that implant-induced bone remodeling post-TKR occurs predominately within the first two post-operative years [67], after which bone turnover tends to stabilize and further changes are mainly attributable to systemic aging [68].

Following the methodology of Minoda et al. [69], changes in BMD were evaluated within three regions of interest (ROIs): a medial ROI, a lateral ROI, and a distal ROI located beneath the implant stem. Each ROI was defined as a square region with 1-cm sides. The medial and lateral ROIs were positioned 1 cm below the inferior surface of the tibial baseplate, while the distal ROI was located 4 cm below the baseplate, as shown in Figure 6.

III. RESULTS

Following implant placement, the tibial baseplate covered 81.4% of the resected tibial plateau surface. This value lies well within the ranges reported in clinical studies for contemporary TKR designs: 80.2–83.8% [50] and 80.6–84.7% [70], indicating that implant sizing and positioning in the present model are representative of clinical practice. Calibration of the strain-adaptive remodeling formulations resulted in a remodeling coefficient $c = 1.68 \times 10^{-4}$ for the modified

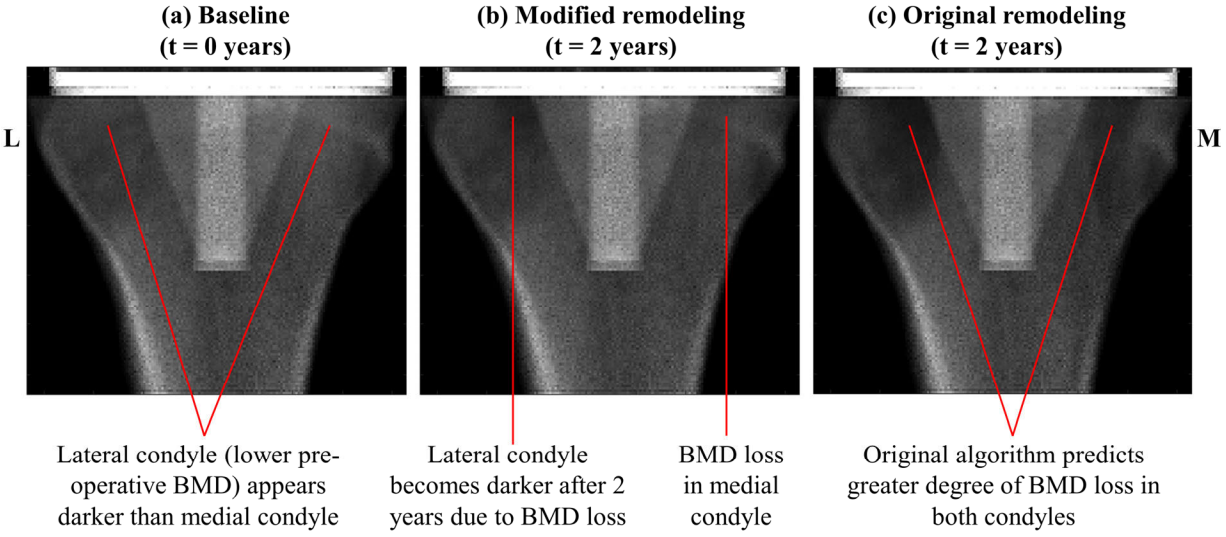


FIGURE 8. Antero-posterior (AP) two-dimensional bone mineral density (BMD) maps of the proximal tibia at (a) baseline (t = 0), (b) after 2 years of remodeling using the modified algorithm, and (c) after 2 years using the original remodeling algorithm. Relative to the modified formulation, the original algorithm predicts greater bone loss beneath the tibial baseplate, while only minimal BMD changes are observed distal to the implant stem. L: lateral. M: medial.

algorithm and $c = 3.50 \times 10^{-3}$ g/[cm^3.mm^2.(J/g).month] for the original formulation. In both formulations, the maximum rate of change in apparent bone density occurred in the early post-operative period and remained below 0.13 g/cm^3/month at all times, in line with clinically observed upper bounds for disuse-related bone loss.

Using the calibrated models, spatial changes in bone apparent density and BMD were evaluated over 2 years. Figure 7 reveals the distribution of ρ in the proximal tibia at baseline (t = 0) and after two years of remodeling using both the modified and original algorithms, while Figure 8 presents AP BMD maps of the proximal tibia at the same time points. At baseline, medial condyle BMD exceeds that of the lateral side, consistent with clinical observations [61], [69]. After two years, both models predict minimal BMD changes in the distal region. In contrast, substantial proximal bone loss is observed beneath the tibial tray. The original strain-adaptive algorithm predicts significantly greater reductions in medial and lateral bone apparent density and BMD than the modified formulation, indicating more extensive stress-shielding-induced resorption. To quantify these spatial patterns, the percentage change in BMD was calculated within the three predefined ROIs (Figure 6). Table 4 summarizes the evolution of BMD loss at 3 months, 6 months, 1 year, and 2 years for both remodeling formulations.

The temporal evolution of percentage change in BMD within the medial, lateral, and distal ROIs over the two-year post-operative period is shown in Figure 9. Average clinical BMD measurements from multiple cemented TKR studies [60], [61], [62], [63], [64], [65] are superimposed for comparison. The data exhibits a relatively widespread distribution due to the heterogeneous nature of patient populations and experimental conditions.

In the medial ROI (Figure 9a), all modified models predict an initial rapid decrease in BMD during the early post-operative period, followed by a pronounced reduction in the rate of loss over time. The original remodeling algorithm, in contrast, predicts significantly greater and continuously increasing bone loss over the entire 2-year period, with values exceeding the range reported by the clinical studies. The modified formulation yields markedly lower proximal bone loss and demonstrates much closer agreement with the clinical data cloud. Extending the duration of assistive-device increased early post-operative bone loss in the medial ROI, with the 2-month and 3-month aid cases predicting larger BMD reductions during the first months. However, the rate of BMD change thereafter remained low, and even in these more conservative loading scenarios the predicted medial BMD loss remained substantially lower than that obtained with the original algorithm and within the scatter of clinical measurements.

Similar trends are observed in the lateral ROI (Figure 9b), although the magnitude of bone loss is consistently lower than that observed medially. The modified remodeling formulations again remain within the observed clinical data spread, while the original algorithm predicts bone resorption that exceeds the upper clinical range. The impact of extended duration of mobility-aids is also most pronounced in the early post-operative period, with little effect on the subsequent rate of density change.

In the distal ROI (Figure 9c), all computational models predict relatively small changes in BMD over the 2-year period. Differences between the original and modified remodeling formulations are less pronounced distally than proximally. This is supported by Figure 10, which displays contour plots for bone density at the mid-coronal plane of the

TABLE 4. Percentage change in BMD over time (3 months, 6 months, 1 year, and 2 years) in the regions of interest defined in Figure 6, predicted using the modified and original strain-adaptive remodeling algorithms.

ROI	Remodeling algorithm	BMD % drop			
		3 months	6 months	1 year	2 years
Medial	Modified	8.09	8.88	10.26	13.33
	Original	13.24	19.38	29.99	38.09
Lateral	Modified	5.66	6.67	8.29	11.81
	Original	9.52	16.45	27.38	35.77
Distal	Modified	1.26	1.25	1.21	1.13
	Original	0.3	0.42	0.55	0.85

proximal tibia at different time points. Even with prolonged use of mobility aids, the numerically predicted distal BMD loss remains below the average clinical values reported in the literature.

IV. DISCUSSION

This study examined time-dependent peri-prosthetic tibial bone remodeling post-implantation using a modified strain-adaptive remodeling framework in which the mechanical stimulus is explicitly coupled to clinically motivated, time-varying loading protocols. By embedding post-operative changes in load magnitude, frequency, and overall activity level directly into the remodeling stimulus, the approach moves beyond the traditional assumption of constant loading and provides a mechanistically more realistic representation of the early rehabilitation period. This, in turn, substantially alters predicted remodeling patterns and yields closer agreement with the magnitude and temporal evolution of bone mineral density (BMD) changes reported clinically than the original strain-adaptive formulation.

A key outcome of the modified framework is its ability to reproduce both the magnitude and temporal progression of clinically observed peri-implant bone resorption. The model predicts greater bone loss in the medial compartment and maintains the characteristic pattern of a rapid decrease in BMD during the early post-operative period, followed by a marked attenuation of remodeling rates thereafter. This response closely mirrors clinical findings, in which most bone loss occurs within the first 3–6 months after surgery, with subsequent changes proceeding more gradually (Figure 9). Mechanistically, this pattern arises from the combination of implant-induced stress shielding and under-loading-related remodeling driven by the reduced effective loading ''dose'' (via load magnitude, frequency, and activity level, as shown in Table 1) immediately after surgery—followed by progressive recovery toward pre-osteoarthritic-pain activity. Because the strain-adaptive law is sensitive to these changes, the model naturally transitions from an initial resorptive regime to a quasi-steady state with reduced net remodeling. This behavior is consistent with clinical evidence demonstrating the beneficial influence of physical activity and exercise on BMD in older adults [71] and postmenopausal populations [72], which constitute a substantial proportion of patients undergoing TKR. In contrast, the original strain-adaptive remodeling algorithm, which assumes constant loading [12], [31], [38], and neglects rehabilitation-related changes in activity, systematically overestimates proximal bone loss and fails to reproduce the observed temporal trends, yielding predictions that remain outside the range reported clinically. These findings suggest that temporal characteristics of loading are not merely secondary modifiers of the remodeling response but represent fundamental determinants of bone adaptation that should be incorporated when simulating post-operative bone remodeling.

Despite the improved agreement achieved with the modified remodeling framework, the clinical data exhibit considerable scatter across regions of interest (ROIs) and studies. Such variability is expected given inter-patient differences in bone quality, alignment, implant choice, and rehabilitation protocols. Studies reporting greater proximal bone loss than predicted by the present model investigated osteoporotic cohorts. Murahashi et al. [64] examined patients with reduced bone quality, which increases the stiffness mismatch between bone and implant, thereby amplifying stress-shielding-induced resorption [73]. Ueyama et al. [60] similarly studied osteoporotic patients, but despite bisphosphonate therapy, which reduces post-TKR bone loss [74], the cohort underwent simultaneous bilateral TKR, a procedure associated with delayed functional recovery and prolonged use of assistive devices [75]. The resulting slower restoration of mechanical loading likely contributed to the greater proximal BMD reductions reported clinically. Likewise, Gazdzik et al. [61] observed pronounced proximal bone loss in patients who relied on mobility aids for three months post-operatively, consistent with the prolonged reduction in effective mechanical stimulus.

Conversely, studies reporting smaller proximal BMD reductions (Jaroma et al. [62] and Soininvaara et al. [65]) permitted immediate full weight-bearing post-surgery. Earlier restoration of near-physiological loading likely limited disuse-related resorption, resulting in less bone loss than in cohorts undergoing more restrictive rehabilitation. Collectively, these comparisons suggest that differences in bone quality and post-operative loading history are major determinants of peri-prosthetic bone adaptation. Capturing this variability will likely require moving beyond population-average loading protocols toward patient-specific loading histories informed by objective measures of activity and assistive device use.

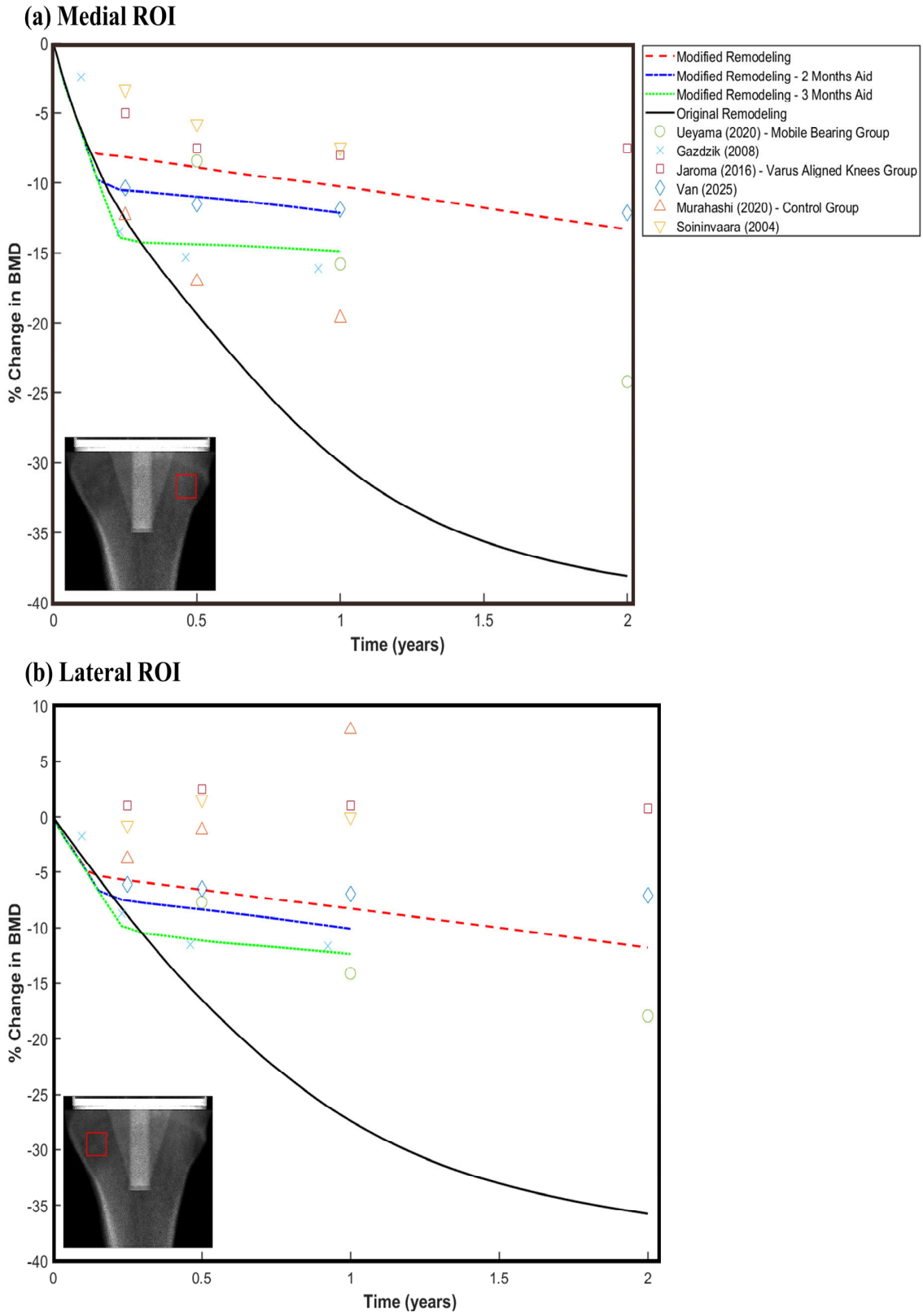


FIGURE 9. Temporal evolution of the percentage change in bone mineral density (BMD) within: (a) medial and (b) lateral distal regions of interest (ROIs) over the post-operative period. Predictions from the modified and original strain-adaptive remodeling formulations are shown, together with 1-year remodeling predictions for two additional modified cases incorporating prolonged use of mobility aids (2 and 3 months). Average clinical BMD data from cemented total knee replacement studies (Ueyama et al. [60], Gazdzik et al. [61], Jaroma et al. [62], Van et al. [63], Murahashi et al. [64], and Soininvaara et al. [65]) are superimposed for comparison.

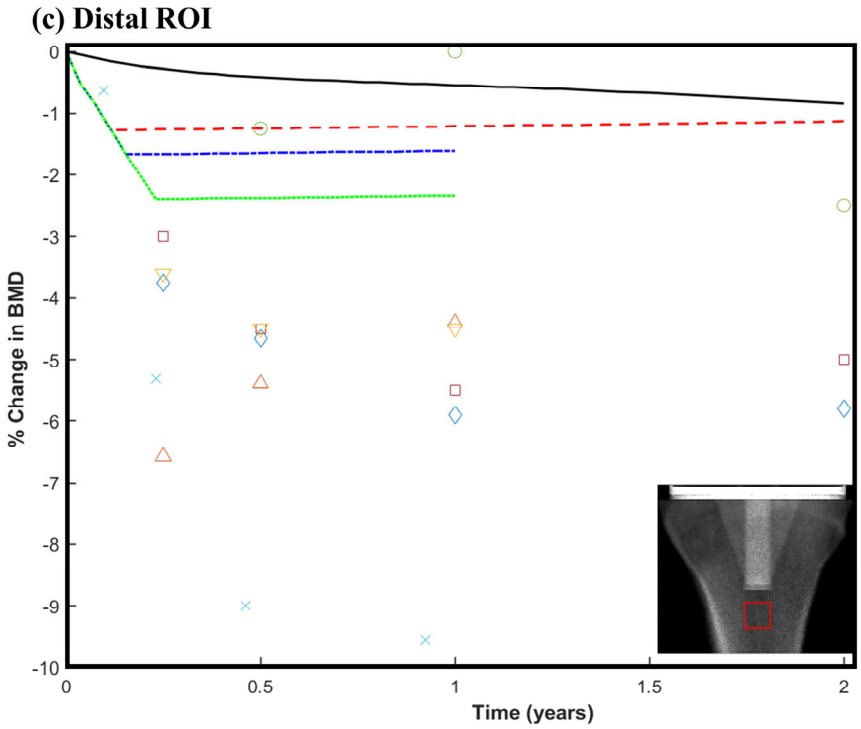


FIGURE 9. *(Continued.)* Temporal evolution of the percentage change in bone mineral density (BMD) within: (a) medial and (b) lateral distal regions of interest (ROIs) over the post-operative period. Predictions from the modified and original strain-adaptive remodeling formulations are shown, together with 1-year remodeling predictions for two additional modified cases incorporating prolonged use of mobility aids (2 and 3 months). Average clinical BMD data from cemented total knee replacement studies (Ueyama et al. [60], Gazdzik et al. [61], Jaroma et al. [62], Van et al. [63], Murahashi et al. [64], and Soininvaara et al. [65]) are superimposed for comparison.

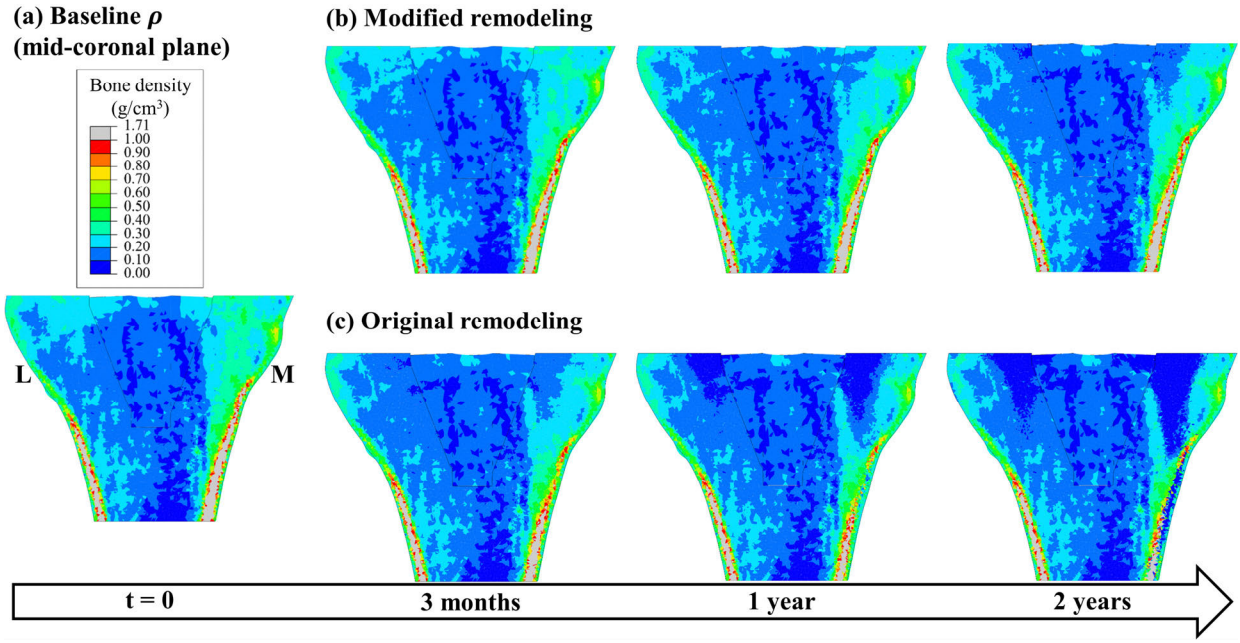


FIGURE 10. Bone density (ρ) contour plots at the mid-coronal plane of the proximal tibia at (a) immediately post-implantation (t = 0), and at t = 3 months, 1 year and 2 years post-implantation for the (b) modified remodeling model and (c) original remodeling model. L: lateral. M: medial.

Among the clinical studies considered, the BMD trends reported by Van et al. [63] are in the closest agreement with the modified remodeling predictions, likely because their cohort exhibited relatively low rates of osteoporosis

and simultaneous bilateral TKR, reducing variability in bone quality and early loading behavior. Nevertheless, the study still reports slightly greater medial bone resorption than predicted numerically. A plausible explanation is the high prevalence of pre-operative varus alignment in patients with knee osteoarthritis [76], which increases medial loading before surgery (up to 75% [58] or nearly the entire joint load [77]) as shown in Figure 11. Surgical correction redistributes loading toward a more balanced medial-lateral distribution, producing a larger reduction in medial mechanical stimulus than represented in the present simulations, which assumed a normally aligned tibia. This simplification likely contributes to the underprediction of medial bone loss. A similar trend is observed when comparing Gazdzik et al. [61] with the three-month mobility-aid simulation, where the clinical bone loss also remains greater, likely because alignment correction was not included. Consistent with this interpretation, extending the modeled use of mobility aids from 6 weeks to 2 months improves agreement with the medial BMD reductions reported in [63], as it mimics to some extent the impact of surgical varus correction on the medial condyle's loading environment, in terms of reduction in mechanical stimulation.

Lateral ROI behavior further supports the role of varus correction and load redistribution (Figure 9b). Clinically, the lateral side generally exhibits minimal BMD reductions (Soininvaara et al. [65]) and, in some cases, bone densification (Jaroma et al. [62] and Murahashi et al. [64]). Moreover, lateral bone loss reported by Van et al. [63] is smaller than that predicted numerically, and similar discrepancies are observed when comparing Gazdzik et al. [61] with the three-month mobility-aid simulation. These findings are consistent with the lateral condyle being relatively underloaded pre-operatively in varus-aligned knees, such that surgical realignment increases its post-operative loading. Consequently, bone resorption is reduced and, in some cohorts, replaced by net densification.

The distal ROI presents a different challenge. Although both the original and modified remodeling formulations underpredict the magnitude of distal BMD loss observed clinically, their temporal responses differ. The original formulation predicts a continuous decline in BMD, whereas the modified algorithm predicts rapid early bone loss, whose magnitude depends on the duration of mobility aid use, followed by gradual recovery as loading normalizes. This trend is consistent with clinical observations of early accelerated resorption followed by stabilization (Jaroma et al. [62]) or partial recovery (Murahashi et al. [64]). However, the persistent underprediction of distal bone loss suggests that the distal mechanical stimulus remains overestimated in the model.

Previous computational studies of cemented TKR [11], [12], [13], [38], [78] have often predicted distal bone densification rather than loss, primarily because fully bonded cement-bone and cement-implant interfaces promote distal load transfer through shear stresses, increasing the local strain energy density beneath the stem [29]. Clinically, however, cement penetration and interface bonding are heterogeneous, and debonding or imperfect interdigitation are common. Consequently, the distal tibia experiences a lower effective mechanical stimulus than that predicted by an idealized perfectly bonded model, favoring bone loss rather than densification.

While the use of distinct resorption and formation remodeling coefficients likely contributes to predicting distal bone loss instead of densification, the remaining discrepancy with clinical observations indicates that further refinements are needed. One possible improvement is the introduction of an asymmetric dead zone with a higher threshold for bone formation than for resorption, allowing moderate increases in mechanical stimulus to suppress bone loss without necessarily inducing net bone formation.

The principal contribution of the proposed framework is not merely improved numerical agreement with clinical BMD measurements, but the ability to explicitly investigate how temporal aspects of loading influence bone adaptation. Unlike the classical strain-adaptive formulation, the proposed model permits remodeling to respond to changes in loading frequency, activity level, and rest periods during rehabilitation. Consequently, clinically relevant questions—such as the influence of prolonged mobility-aid use, delayed rehabilitation, or accelerated return to activity—can now be investigated within the remodeling framework. This extends the application of strain-adaptive remodeling simulations beyond implant assessment to the evaluation of rehabilitation strategies.

Although the proposed framework introduces additional parameters, these quantities are clinically meaningful, readily obtainable from rehabilitation protocols, and possess clear physical interpretation rather than serving as empirical fitting terms. The improved agreement with clinical temporal remodeling trends, together with the ability to evaluate the influence of rehabilitation strategies and loading history, suggests that the modest increase in model complexity is justified by the substantial improvement in physiological realism and predictive capability.

This work is exploratory and subject to several important limitations. First, the tibial model was simplified and did not include structures that significantly modulate load transfer, such as cartilage and menisci in the intact knee, and femoral component and polyethylene insert post-implantation. These components distribute contact stresses, alter joint congruency, and influence the spatial and temporal distributions of loads across the tibial plateau and tray. Their absence implies that the loading field is approximated, which influences the remodeling response. Second, loading was also represented using quasi-static peak compressive loads from three representative activities. Dynamic aspects of gait such as impact peaks, muscle co-contraction, and rapid load rate changes were not captured, nor were the multiple cycles per day that could influence the remodeling stimulus via frequency-dependent effects. Consequently,

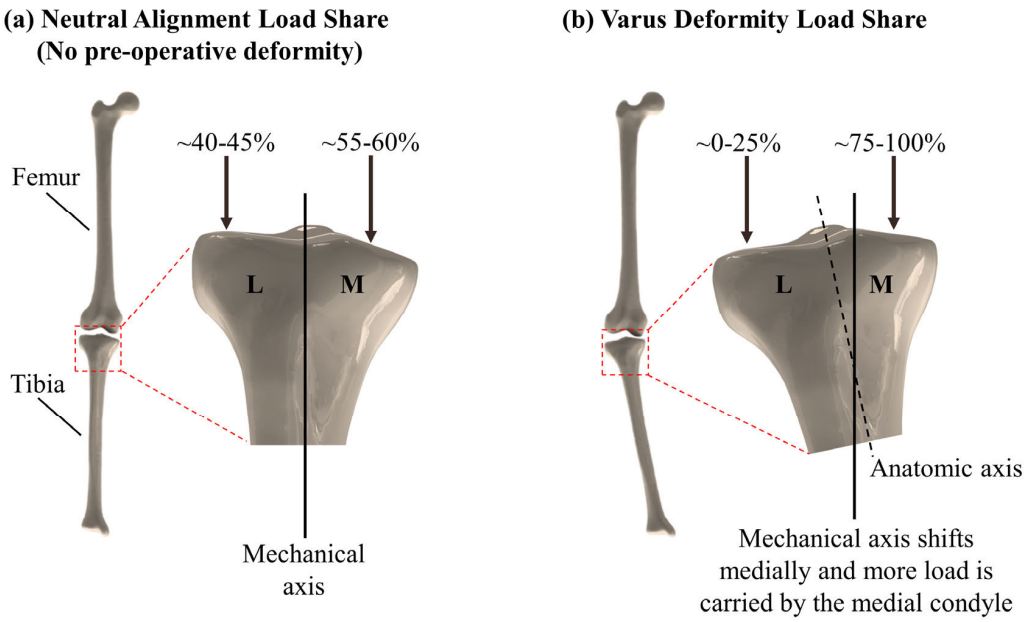


FIGURE 11. Percentage of axial load share in each tibial condyle under (a) neutral alignment with no tibial deformities, where the medial condyle carries between 55 and 60% of the load, while the lateral condyle carries between 40 and 45% of the load, and (b) varus alignment, where the deformity shifts the mechanical axis medially, and the medial condyle carries between 75% and 100% of the load. Following total knee replacement, the neutral tibial alignment is restored, and the medial condyle experiences reduced loading due to implant-induced stress shielding and lower load share, increasing medial bone loss. Conversely, lateral bone loss is relatively less due to the increased load share post-operation. L: lateral. M: medial.

the stimulus used in the model should be viewed as an averaged proxy rather than a fully physiological loading history. Third, as with all CT-based finite element models, limitations in image resolution and voxel discretization may introduce local inaccuracies in the assigned material properties and consequently, the resulting stress and strain fields. Also, the analyses were performed on a single tibial geometry. Although, the donor age (59 years) lies within the typical TKR range, the local cortical and trabecular architecture, bone density, and material properties may not be representative of the broader TKR population. Inter-individual variation in metaphyseal geometry, canal flare, cortical thickness, and trabecular anisotropy can all influence local strains and, hence remodeling trajectories. Extending the approach to a virtual cohort of tibiae with varying morphology and bone quality would be a logical next step to assess the robustness and generalizability of the findings.

Similarly, the pre- and post-operative activity levels and loading protocols adopted here are idealized and may not capture the true distributions of patient behavior. Quantifying daily activity—types of activities, frequency, intensity, and adherence to rehabilitation protocols—is inherently difficult, as is estimating *in-vivo* joint loads, which depend on gait pattern, footwear, surface compliance, and step height. While wearable sensors and instrumented implants offer potential routes to more realistic loading inputs, such data are still not widely available for large TKR cohorts. As a result, the present model necessarily employs standardized, population average assumptions.

Finally, while basing the computational work on a single clinical study could have enabled a more robust comparison between clinical observations and numerical results, many influential variables are often not explicitly reported, including rehabilitation protocols (e.g., use and duration of mobility aids), implant material and thickness, which are parameters known to affect bone response. Given that the present model assumes near population-average bone properties, implant characteristics and a standardized rehabilitation protocol, agreement with the central tendency of clinical data, rather than extreme outcomes, represents the most appropriate benchmark for model validation.

Within this context, the ability of the modified, time dependent strain-adaptive framework to reproduce key qualitative and quantitative trends—greater medial than lateral bone loss, rapid early post-operative resorption followed by stabilization, and sensitivity to changes in activity level and mobility aid use—supports its use as a mechanistic tool for exploring rehabilitation strategies and implant design choices. Nevertheless, the relative influence of the load-related parameters on long-term remodeling predictions has not been quantified independently. Future work should therefore include systematic sensitivity analyses to identify the parameters exerting the greatest influence on predicted bone adaptation and to assess model robustness. Validation against independent longitudinal clinical datasets representing different patient populations [28] and rehabilitation protocols will move the model closer to predictive, individualized simulations capable of informing clinical

decision making and optimizing both surgical strategy and post-operative rehabilitation to preserve periprosthetic bone stock after TKR.

V. CONCLUSION

This study presented a modified strain-adaptive bone remodeling framework for modeling time-dependent tibial bone adaptation post-implantation. By extending the original formulation to incorporate loading frequency, number of loading cycles, and rest periods between loading sessions, together with a clinically motivated time-varying loading protocol, the proposed model provides a more physiologically realistic representation of post-operative rehabilitation and recovery. In addition, the study introduced an approach to calibrate the distinct values of bone resorption and formation remodeling coefficients, as the use of asymmetric remodeling responses better reflects clinically observed bone behavior.

Across medial, lateral, and distal ROIs, these modifications substantially altered predicted remodeling patterns, especially proximally. Two years post implantation, the modified model predicted BMD reductions of 13.33%, 11.81%, and 1.13% in the medial, lateral, and distal ROIs, respectively, versus 38.09%, 35.77%, and 0.85% with the original algorithm. The revised framework thus mitigated the overestimation of proximal bone loss and reproduced key clinical features: marked early post-operative bone loss followed by attenuated resorption and greater medial than lateral bone loss, in line with the central tendency of reported clinical data.

The systematic underprediction of distal bone loss indicates that additional refinements—such as more realistic cement–bone and implant–cement interface modeling and adjusted, possibly asymmetric dead zone thresholds—are needed to fully capture distal remodeling. Overall, the results underscore that activity-related factors and time-varying loading are critical determinants of bone adaptation post-implantation. The proposed framework represents a step toward biologically informed, clinically relevant simulations and provides a foundation for future patient specific, predictive in silico assessment of implant designs, rehabilitation strategies, and long term bone–implant interactions in total knee arthroplasty.

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